Initiation without maintenance: an NMDA receptor–dependent spinal hypothesis for the persistence of hard flaccid syndrome

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[1] *HFS Patient* • [2] *Large language model, Anthropic PBC*

Companion paper: *The sacral splanchnic bypass* (same authors)

Abstract

Background. Hard flaccid syndrome (HFS) usually begins with a mechanical injury to the penis or perineum, yet it rarely remits. The prevailing account — pathological activation of a pelvic/pudendal-hypogastric reflex — explains how symptoms could start but not why they persist for years after an injury that should have healed. Interventions directed at the original injury site, including pudendal decompression and pudendal blockade, are reported to fail. Chronicity, not onset, is the unexplained feature.

Hypothesis. We propose that the peripheral injury initiates but does not maintain HFS, and that what maintains it is a spinal process: an N-methyl-D-aspartate (NMDA) receptor–dependent increase in the excitability and synaptic efficacy of the circuits that drive sympathetic preganglionic neurons projecting to the pelvis. On this account the somato-autonomic limb of the spinal cord undergoes the same form of use-dependent amplification that is well established in the nociceptive limb, and the resulting elevation of pelvic sympathetic outflow becomes at least partly independent of the original afferent source.

Basis. Central sensitization is an established, NMDA receptor–dependent phenomenon in human nociceptive pathways. Somato-sympathetic reflexes are established, spinally organised, and demonstrable in humans; pudendal afferent stimulation can trigger a systemic sympathetic response in people with spinal cord injury. In rodents, NMDA receptors on intermediolateral cell column neurons contribute to resting sympathetic vasomotor tone, and their up-regulation maintains elevated tone in a disease model; noxious peripheral stimulation up-regulates NMDA receptor messenger RNA in sympathetic preganglionic neurons. What has never been shown, in any species, is that these processes combine to produce a self-sustaining elevation of sympathetic outflow after a peripheral nerve injury.

The central objection. If elevated spinal drive were the whole mechanism, then interrupting the sympathetic outflow downstream should abolish the sign — and sympathetic blockade is reported to fail. We treat this objection as the paper's organising problem rather than a footnote, set out five candidate resolutions, and show that three of them demote the hypothesis from a complete account to a stage-specific one. One resolution appears not to have been considered before: a regional pelvic sympathetic block does not reduce adrenal medullary output, so centrally driven humoral catecholamine release would survive any such block.

Tests. The primary test is cheap, non-invasive and available now: standardised quantitative sensory testing against matched controls, looking for enhanced temporal summation, secondary hyperalgesia and aftersensations, with the deficit mapped against the dorsal penile nerve distribution. We also specify a somato-sympathetic reflex gain measurement, a controlled NMDA-antagonist challenge with an objective endpoint and an active placebo, and one prediction that already appears to fail.

Conclusion. The hypothesis is testable, currently untested, and — unlike most accounts of HFS — can be substantially refuted this year with existing equipment. Nothing in this paper should influence clinical management.

Keywords: hard flaccid syndrome; central sensitization; NMDA receptor; sympathetic preganglionic neuron; somato-sympathetic reflex; quantitative sensory testing; ketamine.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. Ketamine and other NMDA antagonists are discussed only as research probes; they are prescription-only, carry substantial psychiatric, urological and dependence risks, and are not proposed as treatment for this condition. Readers should assume the hypothesis is more likely false than true.

1. Introduction

Hard flaccid syndrome (HFS) is an acquired condition in which the penis remains persistently semi-rigid in the flaccid state, usually with penile pain, altered sensation, urinary symptoms, pelvic floor hypertonicity and substantial distress [1, 4, 2, 3]. Goldstein and Komisaruk have proposed that it follows pathological activation of a somato-visceral and/or viscero-visceral reflex — the *pelvic/pudendal-hypogastric reflex* — in which afferent input from the penis, perineum, cauda equina, spinal cord or brain reflexively drives excess sympathetic efferent activity that contracts cavernosal smooth muscle [5, 6].

That account is a plausible description of onset. It is much less clear as a description of persistence. Most patients date their symptoms to a discrete mechanical event — vigorous masturbation, penile elongation practice, or intercourse trauma [1, 2] — yet the condition characteristically continues for years. A stretch injury to a peripheral nerve heals, or at least stops generating an acute inflammatory afferent barrage, on a timescale of weeks to months. Something else is holding the system in its altered state.

The pattern of treatment response points the same way. Interventions directed at the site of the original injury — pudendal nerve decompression surgery, pudendal nerve blockade, intracavernosal botulinum toxin, low-intensity shockwave therapy — are reported to fail with striking consistency **[E]** [8]; symptom resolution was documented in only 2 of the 10 patients whose outcomes were reported in the 2025 systematic review **[B]** [2]. A mechanism that has migrated away from its point of origin would produce exactly that pattern.

This paper proposes and specifies where it may have migrated to. We argue that the maintaining lesion is spinal: an NMDA receptor–dependent increase in the excitability of circuits driving sympathetic preganglionic neurons, analogous to the central sensitization long established in nociceptive pathways but occurring in the somato-autonomic limb. We state the hypothesis formally, grade the evidence, and — because this is where the hypothesis is most vulnerable — devote a full section to the one observation that appears to falsify it.

1.1 Relationship to the companion paper

A companion paper by the same authors examines a different conjecture: that the pathological sympathetic traffic reaches the penis by the sacral splanchnic rather than the hypogastric route [8, 9]. The two hypotheses are not competitors. The present one concerns *what generates* the traffic; that one concerns *which wire carries* it. Both could be true, both could be false, and — as Section 5 shows — the route hypothesis is one of the ways the present hypothesis can survive its central objection.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *hard flaccid syndrome*, *central sensitization*, *somato-sympathetic reflex*, *somato-autonomic reflex*, *sympathetic preganglionic neuron*, *intermediolateral cell column NMDA*, *autonomic dysreflexia*, *quantitative sensory testing* and *ketamine chronic pain*, with backward citation chasing. This was a purposive, non-systematic search; no protocol was registered and no formal risk-of-bias appraisal was applied. Relevant work was probably missed.

Claims are graded as in Table 1. Claims we could not verify against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme used throughout this paper. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence: consistent findings across independent human studies.
[B]	Single human study, human reference text, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Central sensitization is established — in the nociceptive limb

Nociceptor input can trigger a prolonged but reversible increase in the excitability and synaptic efficacy of neurons in central nociceptive pathways **[A]** [10]. This is not a hypothesis; it can be elicited in healthy human volunteers by noxious conditioning stimuli applied to skin, muscle or viscera, and it produces a characteristic and measurable phenotype: dynamic tactile allodynia, secondary punctate or pressure hyperalgesia, aftersensations, and enhanced temporal summation **[A]** [10]. Induction and maintenance depend substantially on glutamate acting at NMDA receptors, with the associated intracellular signalling that follows calcium entry **[A/C]** [10, 11].

Two caveats belong here rather than in the limitations, because they constrain what the primary test below can show. First, Woolf explicitly characterises central sensitization as *reversible* [10]; a condition that almost never remits therefore needs something additional to sustain it, a point we return to in Section 8. Second, and more damaging: although validated measures of sensitization exist, their ability to discriminate sensitization of central from peripheral neurons is unclear **[B]** [VERIFY] [13, 12]. A positive result on the test we propose would therefore be consistent with central sensitization but would not by itself locate the lesion centrally.

3.2 Somato-sympathetic reflexes are established, spinal, and demonstrable in humans

Somatic afferent input produces reflex changes in sympathetic efferent discharge; these somato-sympathetic reflexes were characterised in detail in cat and rat, and comprise a short-latency segmental spinal component and a long-latency component dependent on supraspinal circuitry **[C]** [14, 15]. The organ-level consequences of somatic afferent stimulation on autonomic effectors were subsequently mapped across many systems **[C]** [VERIFY] [16].

The human evidence comes largely from spinal cord injury. In autonomic dysreflexia, sensory stimulation below the lesion — including innocuous cutaneous stimulation — evokes exaggerated sympathetic vasoconstrictor discharge and hypertension, and this is understood as a spinal somato-sympathetic (or viscero-sympathetic) reflex released from supraspinal control, with neural plasticity and circuit reorganisation implicated in its amplification **[B]** [17, 18]. Directly relevant here, pudendal nerve stimulation has been reported to trigger autonomic dysreflexia in humans **[B]** [VERIFY] [19].

This is the closest existing human proof of principle for the reflex that Goldstein and Komisaruk propose: a pudendal afferent volley producing a large sympathetic efferent response. It is also, importantly, a *disanalogy*. Autonomic dysreflexia is primarily a disinhibition phenomenon — loss of descending control over an intact spinal reflex — whereas the present hypothesis proposes amplification of a reflex in a neurologically intact person. That the reflex arc exists and can be pathologically amplified after one kind of insult does not establish that it can be amplified after another **[D]**.

3.3 Sympathetic preganglionic neurons are glutamatergically driven and express NMDA receptors

Sympathetic preganglionic neurons (SPNs) lie in the intermediolateral cell column (IML) of the thoracolumbar cord and receive descending glutamatergic excitation from the rostral ventrolateral medulla **[C]** [20]. Three findings make the IML a candidate site for the process proposed here.

First, spinal NMDA receptors mediate pressor responses evoked from the rostral ventrolateral medulla **[C]** [20], and blockade of excitatory amino acid receptors in the IML reduces baseline sympathetic cardiac drive, implying a *tonic* NMDA-mediated excitatory input at rest **[C]** [VERIFY] [21].

Second — and this is the closest existing analogue of the proposed mechanism — intrathecal NMDA and non-NMDA antagonists applied at T10–T12 each reduce resting sympathetic vasomotor tone, and in experimental endotoxemia the augmented immunoreactivity of the NR1 subunit on IML neurons coincides with the phase in which vasomotor tone is elevated; the authors conclude that up-regulation of NMDA receptors on IML neurons is crucial to the *maintenance* of elevated vasomotor tone **[C]** [VERIFY] [22]. A disease state in which NMDA receptor up-regulation on sympathetic preganglionic neurons maintains pathologically elevated sympathetic outflow therefore already exists in the literature — in a different organ system, a different species and a different provocation.

Third, noxious peripheral stimulation changes NMDA receptor expression in SPNs specifically. After complete Freund's adjuvant injection into the rat forepaw, NMDA receptor and substance P receptor messenger RNA were reported to be up-regulated in preganglionic sympathetic neurons of the intermediolateral nucleus as well as in dorsal horn neurons **[C]** [VERIFY] [23]. Primary sensory fibres containing substance P and glutamate were described as terminating around, and in part directly on, these preganglionic neurons **[C]** [VERIFY] [23].

If that finding is robust, it is the single most important piece of evidence in this paper: it is a direct demonstration, in an animal, that a peripheral noxious insult changes glutamatergic receptor expression in the sympathetic preganglionic population. Because we were unable to retrieve the primary article in full, we flag it and recommend that any reader relying on this argument confirm it first.

3.4 What has never been shown

No study, in any species, has demonstrated that a peripheral nerve injury produces a persistent, NMDA-dependent elevation of sympathetic outflow *to the pelvic viscera* that outlasts the afferent barrage that caused it. The proposition is assembled from adjacent findings. It is an inference, not an observation [D].

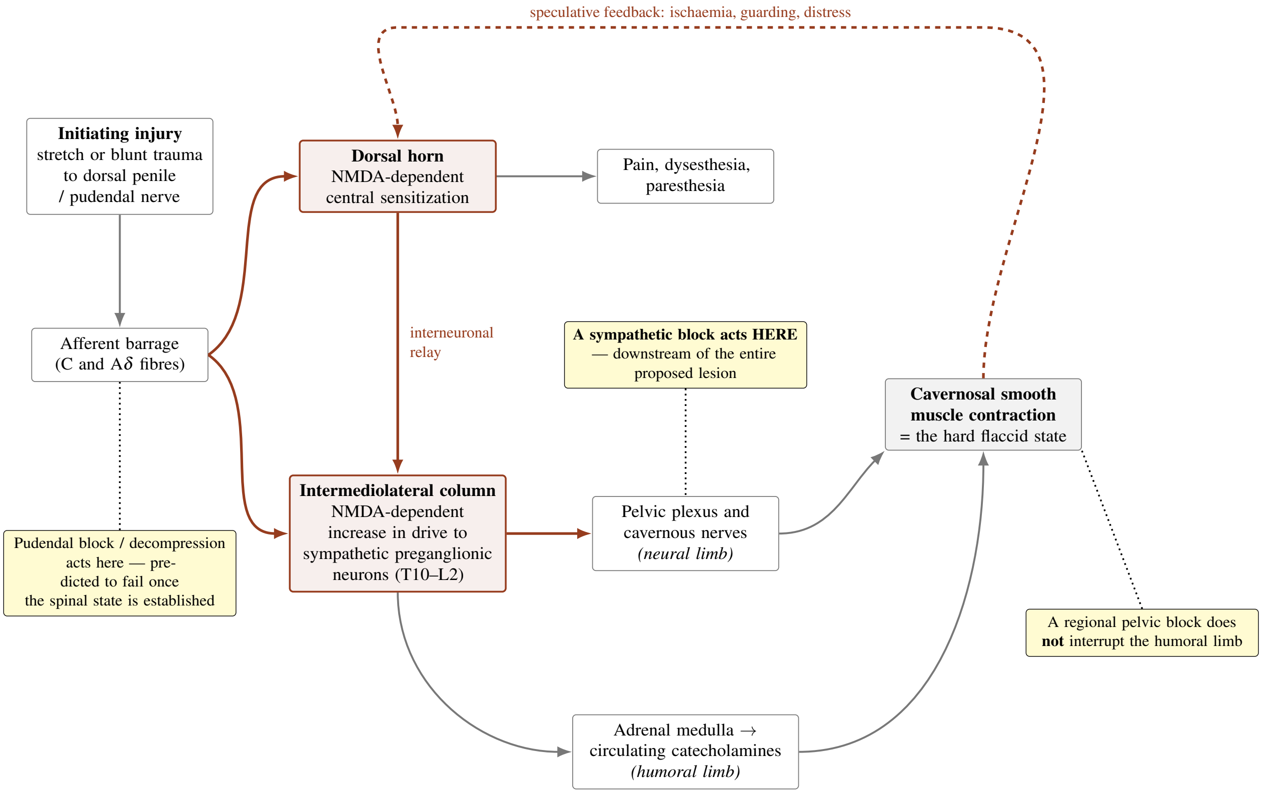

Figure 1. The proposed somato-autonomic sensitization loop, and where each intervention acts relative to it. The initiating injury generates an afferent barrage proposed to sensitise both the nociceptive limb (dorsal horn) and the somato-autonomic limb (drive to sympathetic preganglionic neurons in the intermediolateral column). Elevated preganglionic outflow then contracts cavernosal smooth muscle by two limbs: a neural limb through the pelvic plexus and cavernous nerves, and a humoral limb through adrenal medullary catecholamine release. Note the geometry of the central objection (Section 5): a sympathetic block interrupts the neural limb *downstream* of the entire proposed lesion, so under a pure version of this hypothesis it ought to work — unless the humoral limb, an incorrect route assumption, or effector autonomy intervenes. The dashed feedback arrow is speculative and is included because a self-reinforcing loop is what the hypothesis needs in order to explain chronicity. Diagram is schematic.

4. The hypothesis

Formal statement

H2 (NMDA-dependent central maintenance). In hard flaccid syndrome the peripheral injury initiates but does not maintain the disorder. What maintains it is a spinal process: an NMDA receptor–dependent increase in the excitability and synaptic efficacy of circuits driving sympathetic preganglionic neurons projecting to the pelvis, producing tonically elevated pelvic sympathetic outflow.

H2-strong (autonomous). The spinal state is self-sustaining and independent of ongoing afferent input from the injury. Prediction: abolishing or bypassing all afferent input from the penis and perineum does not reduce the state.

H2-weak (driven). The spinal state requires continuing afferent input to persist — for example from an ectopic generator in remodelled injured axons. Prediction: the state is reduced by measures that silence that input, and the two mechanisms must be targeted together.

H2-null. Maintenance is not central. The elevated tone, if present, is generated peripherally — by the effector tissue itself, by a persisting peripheral generator, or not at all.

H2 rests on three auxiliary assumptions, each stated separately because each is a distinct liability:

- **A1 (limb transfer).** Use-dependent sensitization occurs in the somato-autonomic limb of the spinal cord, not only in the nociceptive limb. Supported by adjacent animal work **[C]** but never demonstrated in humans [DATA NEEDED].

- **A2 (NMDA dependence).** That sensitization is NMDA receptor–dependent at the level of sympathetic preganglionic drive, as it is in the dorsal horn. Partially supported **[C]** [22, 23].
- **A3 (sufficiency).** The resulting increment in pelvic sympathetic outflow is large enough to move cavernosal tone into the symptomatic range. Untested; no dose–response relationship between pelvic sympathetic drive and human cavernosal tone has been characterised [DATA NEEDED].

5. The central objection

5.1 Statement of the objection

If tonically elevated spinal drive is the whole mechanism, then interrupting the sympathetic outflow anywhere downstream of the intermediolateral column should abolish the sign. Sympathetic blockade at the superior hypogastric plexus is reported to fail **[E]** [8]. Figure 1 makes the geometry explicit: a sympathetic block acts downstream of the entire proposed lesion. This is not a peripheral difficulty for H2; it is potentially fatal to it.

We take the objection seriously and set out the candidate resolutions in full. Three of the five demote H2 from a complete account to a stage-specific or partial one, which is itself a substantive conclusion.

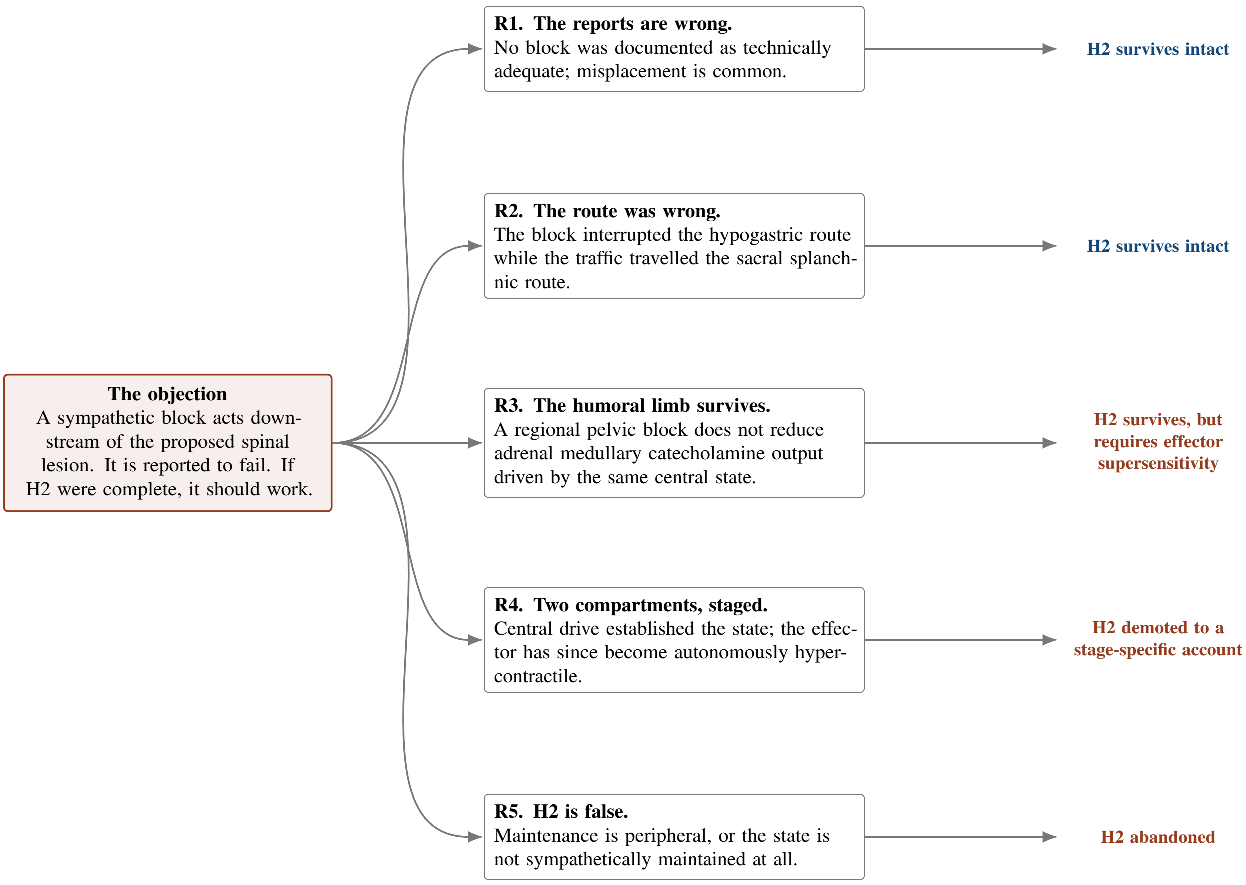

Figure 2. Five candidate resolutions of the central objection, and what each costs the hypothesis. R1 and R2 leave H2 unchanged; R3 preserves it at the cost of an additional assumption about effector sensitivity; R4 preserves the hypothesis only as an account of an earlier disease stage; R5 abandons it. Distinguishing R1 from the rest requires nothing more than documenting block adequacy, which is why we regard that as the most urgent methodological reform in this field.

5.2 R1 — the reports are wrong

Not one reported block failure in HFS is documented as a confirmed, technically adequate block **[E]** [8]. Failure to achieve adequate spread is common in analogous procedures **[B]** [VERIFY] [38]. This is the most parsimonious resolution and it costs H2 nothing. It is also entirely testable, and until it is tested the objection cannot be evaluated at all.

5.3 R2 — the route was wrong

If the pathological traffic reaches the erectile tissue by the sacral splanchnic route rather than the hypogastric nerves, a superior hypogastric plexus block would spare it [9, 8]. This resolution also costs H2 nothing, but it does mean that H2 cannot be evaluated until the route question is settled — the two hypotheses are entangled, and testing either in isolation risks attributing a negative result to the wrong cause.

5.4 R3 — the humoral limb survives the block

This resolution does not appear to have been considered previously and we therefore set it out at slightly greater length.

Sympathetic preganglionic neurons drive not only postganglionic vasomotor fibres but also the adrenal medulla, whose secretory output is preganglionically controlled. A regional sympathetic block placed in the pelvis or on the lumbar chain interrupts the *neural* limb to the erectile tissue. It does not reduce adrenal medullary catecholamine release, and it does not remove circulating catecholamines already in the plasma. If the central state proposed by H2 elevates sympathoadrenal outflow generally rather than only the pelvic projection, then the humoral limb would survive any regional block, and a block could fail while H2 remained true **[D]**.

The cost of this resolution is an auxiliary assumption. Under normal physiological conditions, resting cavernosal tone is thought to be dominated by neurally released noradrenaline at the neuroeffector junction rather than by circulating catecholamines **[D]** [36, 37]. For a humoral contribution to sustain the state alone, the effector must be abnormally sensitive to circulating agonist — which is a separate conjecture [8], and one for which there is no evidence in HFS.

R3 is nonetheless valuable because it generates predictions the other resolutions do not: patients should show measurable systemic sympathoexcitation (elevated resting heart rate, reduced heart rate variability, altered muscle sympathetic nerve activity, or raised plasma catecholamines) relative to matched controls, and systemic α-adrenoceptor blockade should outperform regional sympathetic blockade. Both are testable. The first is heavily confounded by the high reported prevalence of anxiety and depression in this population — 75% in the largest reported series **[B]** [7, 2] — and any such study would need matching or adjustment for anxiety and for medication use [DATA NEEDED].

5.5 R4 — two compartments, staged

If the central state established the disorder but the cavernosal smooth muscle has since become autonomously hypercontractile — through calcium sensitisation, denervation supersensitivity, co-transmitter dominance or pacemaker dysregulation [8] — then no sympathetic block would abolish the state at the time of treatment, and H2 would be true of an earlier stage of the disease only.

This resolution is attractive and dangerous in equal measure. It is attractive because it reconciles reports pointing centrally with reports pointing peripherally, and because it generates the single most actionable recommendation available: **stratify all analyses by symptom duration**. It is dangerous because a staged model can absorb almost any negative result by assigning it to the wrong stage. A staged hypothesis that does not specify, in advance, the duration boundaries at which each mechanism is expected to dominate is close to unfalsifiable. We cannot specify those boundaries — no data exist from which to derive them [DATA NEEDED] — and readers should treat R4 as a research programme rather than an explanation.

5.6 R5 — H2 is false

The remaining possibility is that maintenance is peripheral, or that the premise shared by every sympathetic account of HFS is wrong. Sympathetic outflow to the penis has never been measured in an affected person, and cavernosal tone has never been objectively shown to be elevated. We regard this as the most serious unexamined risk in the field, and it is not specific to H2.

6. Predictions and tests

6.1 The primary test: quantitative sensory testing

The highest-value study available is also the cheapest. Central sensitization has a defined and measurable phenotype — enhanced temporal summation, secondary punctate hyperalgesia, dynamic tactile allodynia, aftersensations **[A]** [10] — and a standardised, multicentre-validated battery exists in the protocol of the German Research Network on Neuropathic Pain, with published age- and sex-stratified reference values and published test–retest and interobserver reliability **[A]** [30, 31]. It requires no new technology, no invasive procedure and no drug.

Prediction P1. Compared with age-matched controls, men with HFS will show a sensitization phenotype on quantitative sensory testing.

Prediction P2. Sensory abnormality will extend *beyond* the anatomical distribution of the dorsal nerve of the penis. This is the more discriminating of the two: a deficit confined to that distribution indicates a peripheral lesion, whereas secondary hyperalgesia in adjacent uninjured territory is a signature of central amplification.

Two constraints on interpretation must be stated in advance rather than discovered afterwards. First, the reference values were established over face, hand and foot [30]; genital and perineal testing would require locally derived control data [DATA NEEDED]. Second, and more fundamentally, current measures cannot reliably distinguish sensitization of central from peripheral neurons **[B]** [VERIFY] [13]. A positive result would therefore be consistent with H2 but would not establish it, and would be equally consistent with a peripheral ectopic generator [8]. A *negative* result is the more informative outcome, because it would substantially undermine H2 and several related conjectures at once.

6.2 The direct test: somato-sympathetic reflex gain

H2 is fundamentally a claim about reflex gain: that a given afferent input now produces a larger sympathetic efferent response than it should. That is measurable in principle. Input–output relationships of the human spinal somato-sympathetic reflex have been characterised using graded electrical stimulation of cutaneous afferents with sympathetic efferent recording **[B]** [18, 17].

Prediction P3. For a standardised, calibrated afferent stimulus applied in a genital or perineal territory, men with HFS will show a larger sympathetic efferent response — a leftward-shifted or steeper input–output curve — than matched controls.

This is the closest thing to a direct test of H2 that we can specify. Its difficulty is the efferent measure. Muscle sympathetic nerve activity recorded by peroneal microneurography indexes the vasoconstrictor outflow to skeletal muscle, not to the pelvic viscera, and the two are not interchangeable **[D]**. Genital sympathetic skin responses have been recorded and normative data exist **[B]** [VERIFY] [33, 34], but a consensus review of neurophysiological assessment in male sexual dysfunction concluded that there is no standardised methodology for these tests and that none is recommended for clinical use, and that corpus cavernosum electromyography remains experimental **[B]** [VERIFY] [32]. Penile surface potentials are further contaminated by cavernous smooth muscle activity **[B]** [35]. Any study using these measures must treat them as unvalidated indices and report their reliability in the study population.

6.3 The pharmacological test: an NMDA-antagonist challenge, done properly

Reported responses to ketamine in HFS are the observation most often cited in support of a central mechanism **[E]** [8]. Section 7 argues that this inference is much weaker than it appears. A properly designed challenge would nonetheless be informative.

Prediction P4. A controlled NMDA-antagonist infusion reduces *objectively measured* resting cavernosal tone relative to an active placebo, and the effect is larger in patients of shorter symptom duration.

Requirements: an objective tone endpoint (see Section 6.5); an *active* placebo producing comparable psychoactive effects, because an inert saline control cannot maintain blinding against a dissociative agent; blinded outcome assessment; and pre-specified stratification by symptom duration.

6.4 A prediction that appears already to fail

The gabapentinoid target $\alpha 2\delta$-1 is not only a calcium channel subunit. It interacts directly with NMDA receptors, and $\alpha 2\delta$-1–bound NMDA receptors have been shown to be required for sympathoexcitatory responses in the hypothalamus; inhibiting $\alpha 2\delta$-1 or disrupting the interaction abolishes the potentiation of NMDA receptor activity and the associated increase in sympathetic outflow **[C]** [24]. The same complex is critically involved in neuropathic pain development and in the therapeutic actions of gabapentin **[C]** [25].

Prediction P5. If HFS is maintained by NMDA receptor–dependent sympathoexcitation, a gabapentinoid at adequate dose should reduce the hard flaccid state and not merely the pain.

The reported clinical observation is the opposite: gabapentin is described as helpful for pain but not for the hard flaccid state **[E]** [8]. If that report is accurate, it is evidence against H2 — weak evidence, because it is anecdotal, because doses are unknown, because the $\alpha 2\delta$-1–NMDA interaction has been characterised in the hypothalamus rather than the spinal IML, and because the sympathoexcitation studied was cardiovascular rather than pelvic. We highlight it because it is the only prediction of H2 for which any relevant observation currently exists, and it points the wrong way. A dose-controlled gabapentinoid challenge with an objective tone endpoint would be a cheap and informative study.

6.5 The endpoint problem

As in the companion paper, every test above is limited by the same gap: **there is no validated objective measure of resting cavernosal tone in humans** [DATA NEEDED]. Candidate approaches — corpus cavernosum electromyography, shear-wave elastography, penile durometry, graded-compression plethysmography — are unvalidated for this purpose, and the one with the longest history is explicitly described as experimental **[B]** [VERIFY] [32]. Until a measure with known test–retest reliability and known variance exists, no sample size can be calculated for any HFS intervention trial, and every reported treatment response rests on patient impression.

The quantitative sensory testing study (P1, P2) is the exception. Its endpoint is sensory, not tonic, and the necessary instrument already exists and is validated [30, 31]. That is the reason we place it first.

Table 2. Predictions of H2 compared with the principal alternatives, and the discriminating value of each observation.

Observation	H2 (central maintenance)	Peripheral ectopic generator	Effector autonomy	Discriminating value
QST sensitization phenotype present	Predicted	Predicted	Not predicted	Low between the first two; a negative result damages both
Sensory abnormality beyond dorsal nerve territory	Predicted	Not predicted	Not predicted	**High**
Elevated somato-sympathetic reflex gain	Predicted	Possible	Not predicted	Moderate–high
Confirmed sympathetic block abolishes the state	Predicted[a]	Predicted	**Not predicted**	**High**
NMDA antagonist reduces objective tone	Predicted	Not predicted	Not predicted	**High**
Response declines with symptom duration	Predicted under R4	Not specified	Predicted (reverse)	Moderate
Gabapentinoid reduces tone, not only pain	Predicted	Not predicted	Not predicted	Moderate
Peripheral interventions fail	Predicted	Not predicted[b]	Predicted	Moderate

[a] Unless resolution R2, R3 or R4 applies — which is precisely why documenting block adequacy and stratifying by duration matter more than any single trial.
[b] A generator distributed along the axon and within the ganglion would not be removed by decompression at a compression point, so this row discriminates less sharply than it appears.

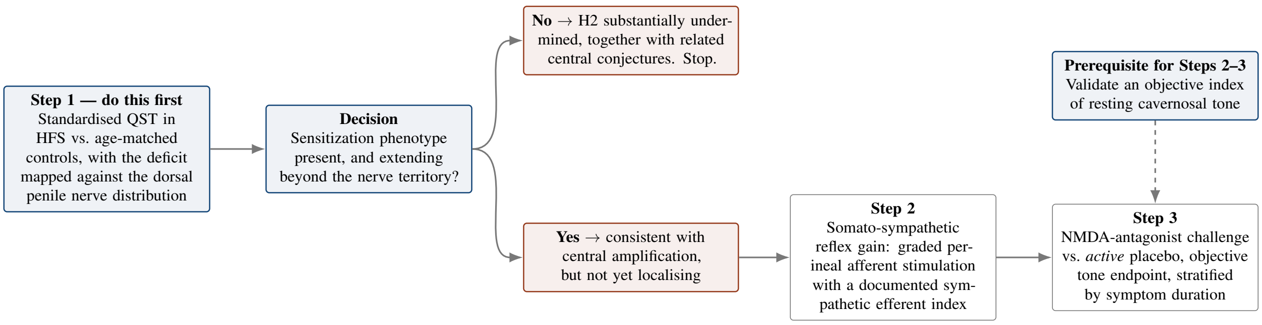

Figure 3. Proposed test sequence, ordered by cost and by how much each step would change the picture. Step 1 requires no new instrument, no drug and no invasive procedure, and a negative result would refute a large family of hypotheses at once; it should therefore precede everything else. Steps 2 and 3 both depend on measurement development that has not been done. All interventions shown are research procedures requiring formal ethical approval and are not proposed as treatment.

7. The ketamine problem

Reported responses to ketamine are the principal clinical motivation for a central hypothesis **[E]** [8]. Four objections make that inference far weaker than it is usually taken to be, and we state them plainly because the temptation to reason from ketamine is strong.

1. **Ketamine is not a selective NMDA probe.** Its hypnotic and analgesic effects involve HCN1 channel blockade and modulation by cholinergic, aminergic and opioid systems in addition to NMDA antagonism **[B/C]** [VERIFY] [29]. More decisively, bespoke ketamine analogues with essentially absent NMDA binding retain profound anaesthetic and analgesic effects, which argues that NMDA receptor antagonism is not the mechanism of ketamine analgesia **[C]** [28]. A response to ketamine therefore does not establish an NMDA-dependent process.

2. **Blinding is close to impossible.** The analgesic dose range for ketamine overlaps its psychotomimetic dose–response profile **[B]** [28]. A dissociative drug evaluated against saline is functionally unblinded, and the endpoint at issue in HFS is a sign the patient assesses by touch and appearance. This is precisely the configuration in which expectancy effects are largest.

3. **The controlled evidence is for pain, and the effect is modest.** In the only randomised double-blind placebo-controlled trial of intravenous ketamine in complex regional pain syndrome, the ketamine group showed a 21.4% reduction in pain score (7.51 ± 0.4 to 6.01 ± 0.6, $p < 0.01$) versus a non-significant 3.1% reduction on placebo **[B]** [26]; a separate randomised trial reported effective and prolonged pain relief in the same condition **[B]** [VERIFY]

[27]. A statistically significant one-fifth reduction in a pain score in a different condition is a thin foundation on which to build a mechanistic claim about smooth muscle tone in another. Effect on an autonomic effector endpoint has not been studied.

4. **Different endpoint, different inference.** Even if ketamine reduced pain in HFS, that would say nothing about the hard flaccid state. The hypothesis at issue concerns the somato-autonomic limb; the trial evidence concerns the nociceptive limb. Conflating them is the error this paper is most at risk of committing.

The appropriate conclusion is that ketamine reports are a reason to design a study, not evidence for a mechanism.

8. Alternative explanations

1. **A persisting peripheral generator.** Remodelling of voltage-gated sodium channel expression in injured and neighbouring afferents can produce ectopic discharge that outlives its cause [8]. This explains chronicity, explains failure of decompression, and would produce a positive quantitative sensory testing result — without any central lesion. It is H2's closest competitor and the two are only distinguished by whether abnormality extends beyond the injured nerve's territory.

2. **Effector autonomy.** Autonomous hypercontractility of cavernosal smooth muscle would explain block failure directly and needs no central account at all [8].

3. **Reversibility mismatch.** Central sensitization is characterised as reversible in principle [10], yet HFS rarely remits. H2 as stated does not explain permanence and requires either a continuing peripheral driver (H2-weak) or an additional maintaining process such as glial activation [8]. This is an internal weakness, not merely an external alternative.

4. **Psychological and behavioural maintenance.** Depression or anxiety was reported in 75% of the largest series, and 45.5% were taking anxiolytics or antidepressants **[B]** [7, 2]; two reported cases improved with psychotropic medication **[B]** [2]. Hypervigilant self-monitoring of a genital sign, with repeated palpation and measurement, is a plausible maintaining mechanism in its own right and is not distinguished from central sensitization by any measurement currently applied. Direction of causation is undetermined: distress is at least as likely to be a consequence of the symptom as a cause of it, and we note that the systematic review's authors reach a similar conclusion cautiously [2].

5. **Heterogeneity.** Only 58.0% of surveyed patients reported onset after a specific incident **[B]** [3]. H2 begins from an initiating injury and can accommodate the remaining 42.0% only by positing subclinical injury — which is absorption, not prediction.

6. **The shared premise may be false.** As in the companion paper: elevated sympathetic outflow to the penis has never been measured in an affected person, and cavernosal tone has never been objectively shown to be raised.

9. Implications

We draw no clinical implications. Three research implications follow.

First, do the sensory study. Quantitative sensory testing against matched controls, with mapping against the dorsal penile nerve distribution, is non-invasive, uses an existing validated protocol, and would inform several competing hypotheses simultaneously. That it has not been done is the most striking gap in this literature.

Second, document block adequacy. Until reported sympathetic block failures are accompanied by imaging confirmation of spread and a physiological index of sympatholysis in a genital territory, the observation on which this paper's central objection rests cannot be evaluated, and neither this hypothesis nor its competitors can be assessed.

Third, stratify by duration. If the staged model (R4) has any merit, pooling patients across symptom durations destroys signal that may already exist in collected data.

10. Limitations

1. **No direct evidence in HFS.** No component of H2 has been demonstrated in an affected person. Sympathetic outflow to the penis has never been measured in one.

2. **The key mechanistic link is assembled, not observed.** No study in any species shows that peripheral nerve injury produces persistent NMDA-dependent elevation of pelvic sympathetic outflow outlasting the afferent barrage.

3. **The most important supporting citation is unverified.** The report of NMDA receptor up-regulation in sympathetic preganglionic neurons after peripheral noxious stimulation [23] could not be retrieved in full and is flagged

accordingly. If it does not support the claim as stated, Section 3.3 weakens substantially.

4. **The human somato-sympathetic evidence is disanalogous.** Autonomic dysreflexia is a disinhibition phenomenon after spinal cord injury; the proposed mechanism is amplification in a neurologically intact person.
5. **Species and preparation.** The intermediolateral column findings are from anaesthetised rats, in cardiovascular rather than pelvic sympathetic outflow, using intrathecal pharmacology and an endotoxemia model with no evident relationship to nerve injury.
6. **The central objection may be unanswerable.** Three of the five resolutions cost the hypothesis something; one (R4) risks unfalsifiability, and we cannot specify the stage boundaries that would make it testable.
7. **The primary test cannot localise.** Available measures of sensitization do not reliably distinguish central from peripheral neurons [13], so a positive result would be consistent with a competing hypothesis.
8. **No endpoint for the tone measurement.** As in the companion paper, no validated objective measure of resting cavernosal tone exists, so no power calculation is possible [DATA NEEDED].
9. **Non-systematic search**, with several citations verified only against secondary sources; all such are marked [VERIFY].
10. **Confirmation risk.** This paper was written to develop a specific conjecture. We have tried to counter that by grading every claim, by giving the central objection its own section, and by reporting a prediction that appears already to fail; the reader should still discount accordingly.

11. Refutation criteria

- **Substantially refuted.** Quantitative sensory testing against matched controls shows no sensitization phenotype in men with HFS.
- **Substantially refuted.** Sensory abnormality maps precisely onto the distribution of the dorsal nerve of the penis with no extraterritorial spread, indicating a purely peripheral lesion.
- **Substantially refuted.** A controlled NMDA-antagonist challenge against an active placebo, with an objective tone endpoint, produces no reduction in resting tone.
- **Weakened.** Somato-sympathetic reflex input–output relationships in HFS are indistinguishable from matched controls.
- **Weakened.** A dose-controlled gabapentinoid challenge reduces pain but leaves objectively measured resting tone unchanged.
- **Rendered moot.** Direct measurement shows sympathetic outflow to the penis is normal or reduced in symptomatic patients, or that resting cavernosal tone is not objectively elevated.

12. Conclusion

The prevailing account of hard flaccid syndrome explains onset better than it explains persistence, and persistence is the feature that makes the condition what it is. We have proposed that the maintaining lesion is spinal — an NMDA receptor–dependent amplification of the circuits driving sympathetic preganglionic neurons — and have tried to state it precisely enough to be attacked.

The hypothesis has real support in adjacent literature: central sensitization is established and NMDA-dependent; somato-sympathetic reflexes are established and demonstrable in humans; NMDA receptors on sympathetic preganglionic neurons contribute to resting sympathetic tone and their up-regulation maintains elevated tone in at least one animal disease model; and peripheral noxious stimulation has been reported to up-regulate NMDA receptor expression in those same neurons. It also has a serious problem, which we have not solved: a sympathetic block acts downstream of everything the hypothesis proposes, and is reported to fail. Three of the five available resolutions cost the hypothesis something, and one of them — the staged model — risks making it unfalsifiable unless stage boundaries are specified in advance, which no current data permit.

Two things follow. The first is that the most useful study in this field right now is not a mechanistic one but a sensory one: standardised quantitative sensory testing against matched controls, mapped against the dorsal penile nerve distribution, requiring no new technology and capable of damaging several hypotheses at once. The second is that reported treatment failures are worthless as evidence unless technical adequacy is documented. Both are cheap. Neither has been done. Until they are, hard flaccid syndrome will keep accumulating explanations it has no way to discard, and this paper will have contributed one more.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Every citation, and particularly those marked [VERIFY], should be checked against the primary source first.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [8] from which this conjecture is drawn; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. Ketamine, gabapentinoids, sodium channel blockers and sympathetic nerve blocks are prescription-only interventions carrying real risk, appropriate only under qualified medical supervision and, in this condition, only within an approved research protocol.

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